Supplementary Material for "AAFV-Net: Guarded Three-Way Age-Aware Face Verification With Empirically Bounded Random False-Accept Risk"

Anurag Nayak, *Graduate Student Member, IEEE,* and [Co-Authors]

S-I. Complete 26-Variant Calibration Ablation Audit

Table I lists all 26 frozen ablation variants and comparator architectures audited on the MORPH `calibration` split (338,871 total pairs; $\mathcal{D}_{\text{hard}}$ $N = 35{,}769$, except VLM $N = 1{,}500$), paired with decision rates on the label-free calibration common subset ($\beta = 1.0\%$, `decision_table_calibration.csv`).

S-II. Prior-Adjusted Calibration Reliability by Age-Gap Bin

Table II reproduces `audit/reliability_by_bin.csv`, verifying calibration alignment between mean predicted probability and observed genuine frequency on `train_val` across age-gap bins G1–G4 (with G5–G6 omitted due to < 20 genuine `train_val` samples).

S-III. IEEE TIFS Deep Learning Reproducibility Checklist & Checksums

- **Backbone Extractors (Frozen):** InsightFace `buffalo_l` ArcFace (ResNet-100, 112×112), EfficientNet-B0, VGG-Face, AlexNet LPIPS, analytical SSIM, StyleGAN2 SAM (`sam_ffhq_aging.pt`), and DEX age estimator.
- **Image Quality Normalization (`quality.py`):** Five raw features (`sharp`, `illum`, `contrast`, `noise` inverted, `detail`) min-max scaled via 5th/95th `train_fit` percentiles; pair quality $Q = 0.5(Q_1 + Q_2)$; empirical 5th percentile $Q_{\min} = 0.294763$.
- **Fusion MLP (`fusion.py`, `p6_train.py`):** 5-seed ensemble (947 weights/seed: 3×6 linear heads $+929$ MLP $11 \to 32 \to 16 \to 1$, ReLU, Dropout 0.10), trained with AdamW (lr $= 0.003$, wd $= 0.01$, batch size 2,048), inverse gap-bin frequency weights, SAM branch masking $p_{\text{mask}} = 0.12$, early stopping patience 10 ($\Delta < 10^{-5}$; stopped at epochs 30, 36, 45, 47, 27).
- **Baseline Protocols (`p11_siamese.py`, `p11_vlm.py`):** Siamese CNN (`resnet18`, 128×128, 6 epochs, stopped epoch 4, best val log loss 0.1950); Siamese ViT (`vit_base_patch16_224.augreg_in21k_ft_in1k`, 224×224, lr $= 2 \times 10^{-5}$, 3 epochs, best val log loss 0.1285); Qwen2-VL-2B (4-bit LoRA $r = 16, \alpha = 16$, 224×224, 8,000 balanced pairs, 1 epoch, scored strictly on calibration common subset and FG-NET).
- **Cryptographic Sealing:** Look 1 `thresholds.json` LF SHA-256 `4b7af9772d209ce6315d36f36df6b191728c830159f04355e7a2238fd9732cfe` (`test_access.log` Look 1: `2026-11-05T21:42:09Z`); Look 2 `ablations/variants/full/frozen.pkl` SHA-256 `a47c916bc819379f9fa5a35b29fdc5ecf3a09f092fbe1cb2c4fbf3347f0035b9` (`test_access.log` Look 2: `2026-11-07T19:45:00Z`).

TABLE I
FULL 26-VARIANT CALIBRATION PERFORMANCE AND SUBSET DECISION OUTCOMES ($\beta = 1.0\%$). ZERO FALSE ACCEPTS OCCURRED ON CALIBRATION ACROSS EVERY VARIANT.

Variant ID	**Group**	T_{acc}	ρ	**Full Cal. Accept (%)**	**AUC ($\mathcal{D}_{\text{random}}$)**	**AUC ($\mathcal{D}_{\text{hard}}$)**	**TPR@10^{-3} ($\mathcal{D}_{\text{hard}}$)**	**Subset Accept (%)**	**Subset Hard Reject (%)**
full	main	0.9564	0.20	73.26	0.9872	0.9981	0.9732	74.13	98.20
arcface_only	baseline	0.8497	0.20	87.65	0.9920	0.9802	0.9689	87.20	0.00
arcface_only_noguard	baseline	0.8497	0.00	92.00	0.9920	0.9802	0.9689	91.60	0.00
direct5_logreg	baseline	0.9134	0.20	84.86	0.9920	0.9805	0.9709	84.27	0.00
direct_only_mlp	branch	1.0419	0.20	0.00	0.9868	0.9982	0.9714	0.00	98.47
sam_only_mlp	branch	1.0006	0.20	0.00	0.9900	0.9826	0.7653	0.00	44.80
drop_arc	scores	0.9944	0.20	0.81	0.9714	0.9134	0.1384	0.73	17.87
drop_eff	scores	0.9781	0.20	80.81	0.9870	0.9984	0.9723	81.27	97.60
drop_vgg	scores	0.9759	0.20	73.07	0.9871	0.9980	0.9731	73.93	97.20
drop_lpips	scores	0.9995	0.20	62.37	0.9876	0.9982	0.9731	62.87	97.47
drop_ssim	scores	0.9890	0.20	71.38	0.9874	0.9980	0.9729	72.80	97.40
arc_only_scores	scores	1.0078	0.20	0.00	0.9874	0.9984	0.9714	0.00	97.73
fusion_logreg_no_mlp	fusion	0.8690	0.20	63.98	0.9914	0.9807	0.9675	64.67	0.00
no_context	fusion	1.0422	0.20	0.00	0.9871	0.9978	0.9722	0.00	98.07
no_reliability	fusion	0.9891	0.20	71.70	0.9874	0.9982	0.9732	73.27	97.80
no_branch_masking	fusion	0.5928	0.35	3.44	0.9874	0.9984	0.9732	3.60	98.40
no_pair_weights	fusion	0.9854	0.20	72.74	0.9873	0.9984	0.9730	74.40	97.33
agg_fwd_only	aggregation	0.9542	0.20	73.89	0.9875	0.9982	0.9729	75.07	98.27
agg_no_shared_targets	aggregation	0.9970	0.20	83.61	0.9873	0.9983	0.9736	84.80	98.27
agg_shared_only	aggregation	1.0241	0.20	0.00	0.9872	0.9983	0.9729	0.00	98.13
guard_none	guards	1.0054	0.00	0.00	0.9872	0.9981	0.9733	0.00	98.20
guard_quality_only	guards	1.0054	0.00	0.00	0.9872	0.9981	0.9733	0.00	98.20
guard_quality_reliability	guards	1.0054	0.00	0.00	0.9872	0.9981	0.9733	0.00	98.20
baseline_siamese_cnn	external	0.9815	0.20	0.58	0.9867	0.9530	0.3109	0.27	10.87
baseline_siamese_vit	external	1.0014	0.20	0.00	0.9884	0.9724	0.6050	0.00	12.20
baseline_vlm	external	0.9977	0.20	0.00	0.9802	0.9063	0.0540	0.00	10.20

TABLE II
CALIBRATION PROBABILITY RELIABILITY CHECK ACROSS AGE-GAP BINS (`RELIABILITY_BY_BIN.CSV`).

Bin	N	**Same**	**Mean P**	**Observed**	**Abs. Dev.**	**Status**
G1	108,474	23,304	0.5006	0.5014	0.0008	Calibrated
G2	76,785	605	0.0228	0.0217	0.0011	Calibrated
G3	60,398	177	0.0208	0.0198	0.0010	Calibrated
G4	47,570	44	0.0080	0.0154	0.0074	Calibrated
G5	53,731	78	–	–	–	skipped: $<$20 same-person pairs in train_val
G6	16,689	35	–	–	–	skipped: $<$20 same-person pairs in train_val